

A VECTOR METHOD FOR SOLVING LINEAR EQUATIONS AND INVERTING MATRICES

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Introduction. In a recent article E. W. Purcell (1) presented a vector method for finding solutions of simultaneous linear equations; the method he described does not always work, however, since division by zero may occur. In this paper we remedy this defect, add a simplification and extend the method to matrix inversion. Also we present a simple scheme for solving equations with paper and pencil or desk computer, having the advantage over Cramer's method, for instance, that one's work may be checked at each stage of the computation. Some advantages and disadvantages of the method of matrix inversion are discussed: An important advantage from the standpoint of digital machine calculation is that it permits rescaling at each stage which makes the method less sensitive to round off errors than several other methods; a disadvantage is its relative slowness, requiring roughly $5n^3$ operations to invert an $n \times n$ matrix. In another paper we shall show how this can be obtained.

1. Vector interpretation of solving systems of equations. Consider a system of equations

$$\begin{array}{l} a_{11}x_1 + \cdots + a_{1n}x_n + b_1 = 0 \\ \vdots \\ a_{nn}x_1 + \cdots + a_{nn}x_n + b_n = 0 \end{array} \quad (1)$$

This system defines the set of n row vectors in $(n + 1)$ dimensional Euclidean space

$$E_i = (a_{i1}, a_{i2}, \dots, a_{in}, b_i) \quad i = 1, \dots, n \quad (2)$$

The system (1) may then be interpreted as a statement that the vector $(x_1, \dots, x_n, 1)$ is orthogonal to each of the E_i . The problem of solving (1) may then be stated as that of finding such a vector which is orthogonal to each of the row vectors E_i ; more generally the problem is to find a basis for vectors of this type which are orthogonal to all the E_i .

We shall present a method for constructing such a basis, the method depending upon the following lemma.

2. Lemma. Let X be a subspace of a vector space V , $u_1, \dots, u_m$ a basis for X and E a vector in V which is not orthogonal to u_1 , $E \cdot u_1 \neq 0$. Then the vectors

$$\begin{aligned} v_1 &= u_1 \langle u_2 \cdot E \rangle - u_2 \langle u_1 \cdot E \rangle \\ v_2 &= u_1 \langle u_3 \cdot E \rangle - u_3 \langle u_1 \cdot E \rangle \\ &\vdots \\ v_{m-1} &= u_1 \langle u_m \cdot E \rangle - u_m \langle u_1 \cdot E \rangle \end{aligned}$$

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are (1) orthogonal to E , (2) linearly independent, and (3) form a basis for the subspace Y of X which is orthogonal to E .

Proof: By direction computation $E \cdot v_j = (E \cdot u_1)(u_{j+1} \cdot E) - (E \cdot u_{j+1})(u_1 \cdot E) = 0$, $j = 1, \dots, m-1$. Let

$$v_m = u_1.$$

Let us write out the linear transformation defined in the above equations as a matrix equation

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{pmatrix} = \begin{pmatrix} u_2 \cdot E, & -u_1 \cdot E, & 0, & \dots & 0, & 0 \\ u_3 \cdot E, & 0, & -u_1 \cdot E, & 0, & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ u_m \cdot E, & 0, & 0, & \dots & 0, & -u_1 \cdot E \\ 1, & 0, & 0, & \dots & 0, & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ \vdots \\ u_m \end{pmatrix}$$

It can be seen immediately that the transformation matrix is nonsingular, for expanding the determinant by the last row we obtain $\pm(u_1 \cdot E)^m$, and $u_1 \cdot E$ is not zero by hypothesis. Hence the inverse transformation exists so that $u_1, \dots, u_m$ can be expressed as linear combination of the vectors $v_1, \dots, v_m$. Thus $v_1, \dots, v_m$ constitute a basis for X . It easily follows that $v_1, \dots, v_{m-1}$ form a basis for the subspace Y of X orthogonal to E , and Y is $(m-1)$ dimensional, q.e.d.

3. Construction of the solution by the vector method. Referring to equations (1) and (2), we start with the following basis of unit vectors for Euclidean $(n+1)$ space: $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_{n+1} = (0, 0, \dots, 1)$. We form the inner products $e_1 E_1, e_2 E_1, \dots, e_{n+1} E_1$ and assume without loss of generality that $e_1 E_1 \neq 0$. Then by the above lemma the following set

$$f_i = (e_{i+1} E_1) e_i - (e_i E_1) e_{i+1}, \quad i = 1, \dots, n$$

of n vectors are orthogonal to E_1 and in fact form a basis for the n -dimensional subspace orthogonal to E_1 . Now we form the inner products $f_1 E_2, f_2 E_2, \dots, f_n E_2$. If all of these are zero then E_1 and E_2 are linearly dependent, the f 's automatically satisfy E_2 and we may pass on directly to E_3 . If one of these products is not zero, say $f_1 E_2$, then the $n-1$ vectors

$$g_j = (f_{j+1} E_2) f_1 - (f_1 E_2) f_{j+1} \quad j = 1, \dots, n-1$$

are orthogonal to E_2 and, being linear combinations of the f_i , are also orthogonal to E_1 and form a basis for the orthogonal complement of the subspace spanned by E_1 and E_2 .

Proceeding inductively, we suppose that

$$g_1, \dots, g_{n+1-k}$$

is a basis for the subspace in E^{n+1} Euclidean $(n+1)$ space orthogonal to the subspace spanned by

$$E_1, \dots, E_k \quad 1 \leq k < n.$$

We proceed to E_{k+1} ; if $g_i E_{k+1} = 0$ for all g_i then we already have a basis orthogonal to the subspace generated by $E_1, \dots, E_{k+1}$. If, however, say $g_1 \cdot E_{k+1} \neq 0$ then

$$h_r = (g_{r+1} \cdot E_{k+1})g_1 - (g_1 \cdot E_{k+1})g_{r+1}, \dots, r = 1, \dots, n - k$$

forms a basis orthogonal to the subspace spanned by $E_1, \dots, E_{k+1}$. Thus, proceeding inductively we obtain a basis for the subspace orthogonal to all the E_i , or we exhaust the n -dimensional subspace before we exhaust the E_i , in which case there is no solution. Thus either we obtain a basis for all solutions of the equations (1) or else no solution exists. We will now consider a numerical example.

4. A scheme for systematizing the solution of linear equations. We shall now outline a "*rote*" method for solving equations, which can easily be carried out with paper and pencil with or without a desk computer for not too large systems of equations. We shall work out an example for 5 equations in 5 unknowns. Consider then the system

$$\begin{aligned} x_1 + 2x_2 & \quad + 2x_4 + x_5 + 0 = 0 \\ 3x_1 + 2x_2 + x_3 + 2x_4 + 3x_5 + 0 & = 0 \\ x_1 + 3x_2 + 5x_3 + x_4 + x_5 + 2 & = 0 \\ x_1 & \quad - x_4 + x_5 + 1 = 0 \\ 2x_1 + 2x_2 + x_3 + x_4 + 3x_5 - 1 & = 0 \end{aligned} \tag{3}$$

Table I will be almost self-explanatory:

1. First write down the row vectors $E_1, \dots, E_5$ defined by (3), underneath these, the unit vectors $e_1, \dots, e_5$;
2. Then in a single column just to the right of each e vector separated from it by bold lines we record the products of E_1 with each e vector.
3. We encircle any one of these numbers which is not zero; in this example we choose $e_1 \cdot E_1$, and *change its sign*.
4. We then compute a new vector f_1 by multiplying e_1 by the number $(e_2 \cdot E_1)$ opposite e_2 and adding this vector to the vector e_2 multiplied by the (new) number opposite e_1 ,

$$\begin{array}{ccc} (1, 0, 0, 0, 0) & \begin{array}{c} \nearrow 1^- \\ \searrow 2 \end{array} & \\ (0, 1, 0, 0, 0) & & \end{array}$$

We use this "criss-cross" method to compute the remaining f 's which are recorded directly below the e 's (cf. adjacent formulas for the f 's).

5. We form the products of E_2 with the f 's and record these in the column just adjacent to the f 's.
6. We encircle one of these which is nonzero (in this example we choose $f_2 \cdot E_2$) and *change its sign*.

TABLE I
Scheme for Solving a System of Five Equations in Five Unknowns

E_1	1	2	0	2	1	0
E_2	3	2	1	2	3	0
E_3	1	3	5	1	1	2
E_4	1	0	0	-1	1	1
E_5	2	2	1	1	3	-1

e_1	1	0	0	0	0	0	① ⁻	(= $e_1 \cdot E_1$)	
e_2	0	1	0	0	0	0	2	(= $e_2 \cdot E_1$)	
e_3	0	0	1	0	0	0	0	(= $e_3 \cdot E_1$)	
e_4	0	0	0	1	0	0	2	(= $e_4 \cdot E_1$)	
e_5	0	0	0	0	1	0	1	(= $e_5 \cdot E_1$)	
e_6	0	0	0	0	0	1	0	(= $e_6 \cdot E_1$)	

f_1	2	-1	0	0	0	0	4	(= $f_1 \cdot E_2$)	$f_1 = e_1(e_2 \cdot E_1) - e_2(e_1 \cdot E_1)$
f_2	0	0	-1	0	0	0	-① ⁺	(= $f_2 \cdot E_2$)	$f_2 = e_1(e_3 \cdot E_1) - e_3(e_1 \cdot E_1)$
f_3	2	0	0	-1	0	0	4	(= $f_3 \cdot E_2$)	$f_3 = e_1(e_4 \cdot E_1) - e_4(e_1 \cdot E_1)$
f_4	1	0	0	0	-1	0	0	(= $f_4 \cdot E_2$)	$f_4 = e_1(e_5 \cdot E_1) - e_5(e_1 \cdot E_1)$
f_5	0	0	0	0	0	-1	0	(= $f_5 \cdot E_2$)	$f_5 = e_1(e_6 \cdot E_1) - e_6(e_1 \cdot E_1)$

g_1	2	-1	-4	0	0	0	-21	(= $g_1 \cdot E_3$)	$g_1 = f_2(f_1 \cdot E_2) - f_1(f_2 \cdot E_2)$
g_2	2	0	-4	-1	0	0	-19	(= $g_2 \cdot E_3$)	$g_2 = f_2(f_3 \cdot E_2) - f_3(f_2 \cdot E_2)$
g_3	1	0	0	0	-1	0	0	(= $g_3 \cdot E_3$)	$g_3 = f_2(f_4 \cdot E_2) - f_4(f_2 \cdot E_2)$
g_4	0	0	0	0	0	-1	-② ⁺	(= $g_4 \cdot E_3$)	$g_4 = f_2(f_5 \cdot E_2) - f_5(f_2 \cdot E_2)$

h_1	4	-2	-8	0	0	21	+②⑤ ⁻	(= $h_1 \cdot E_4$)	$h_1 = g_4(g_1 \cdot E_3) - g_1(g_4 \cdot E_3)$
h_2	4	0	-8	-2	0	19	25	(= $h_2 \cdot E_4$)	$h_2 = g_4(g_2 \cdot E_3) - g_2(g_4 \cdot E_3)$
h_3	2	0	0	0	-2	0	0	(= $h_3 \cdot E_4$)	$h_3 = g_4(g_3 \cdot E_3) - g_3(g_4 \cdot E_3)$

i_1	0	-2	0	2	0	2	-④ ⁺	(= $i_1 \cdot E_5$)	$i_1 = h_1(h_2 \cdot E_4) - h_2(h_1 \cdot E_4) (1/25)$
i_2	-2	0	0	0	2	0	2	(= $i_2 \cdot E_5$)	$i_2 = h_1(h_3 \cdot E_4) - h_3(h_1 \cdot E_4) (1/25)$

j	-4	-2	0	2	4	2			$j_1 = i_2(i_1 \cdot E_5) - i_1(i_2 \cdot E_5) (1/2)$
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Answer: $x_1 = -4/2 = -2$

$x_2 = -2/2 = -1$

$x_3 = 0/2 = 0$

$x_4 = 2/2 = 1$

$x_5 = 4/2 = 2$

(We divided i_1 , i_2 by 25 and j_1 by 2 merely to reduce the size of the numbers.)

7. We then use the criss-cross method to form the g 's (cf. adjacent formulas).
8. And so on until we arrive at j .
9. Finally, we divide j by its last component, which will always be nonzero if there is a solution to the original system of equations, to give the solution

5. Solving matrix equations. We shall indicate by means of an example how the method is extended to inverting matrices. Consider the matrix equation

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x & y & z \\ u & v & w \\ p & q & r \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4)$$

Considering just the first column of each side of the equation, we have the equations

$$a_{11}x + a_{12}y + a_{13}z - 1 = 0 \quad (5)$$

$$a_{21}x + a_{22}y + a_{23}z - 0 = 0 \quad (6)$$

$$a_{31}x + a_{32}y + a_{33}z - 0 = 0 \quad (7)$$

which, considering x, y, z as unknowns, we see to be a special case of equation (1). We could therefore solve this equation by the method outlined in article 1, but it may be done even more simply by taking advantage of the zeros appearing in the first column of the identity matrix of equation (4); we interpret equations (6) and (7) as stating that the 3-vector (x, y, z) is orthogonal to the row vectors $E_2 = (a_{21}, a_{22}, a_{23})$ and $E_3 = (a_{31}, a_{32}, a_{33})$. Then consider unit vectors $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$. We now form $e_1 \cdot E_3$, $e_2 \cdot E_3$, $e_3 \cdot E_3$. At least one of these products will be non-zero unless $E_3 = (0, 0, 0)$, which case we rule out. Suppose then that $e_1 \cdot E_3 \neq 0$. We form

$$f_1 = (e_2 \cdot E_3)e_1 - (e_1 \cdot E_3)e_2$$

$$f_2 = (e_3 \cdot E_3)e_1 - (e_1 \cdot E_3)e_3.$$

Then $f_1 \cdot E_3 = f_2 \cdot E_3 = 0$. Next we form $f_1 \cdot E_2$ and $f_2 \cdot E_2$. Again not both of these will be zero and $g = (f_1 E_2) f_2 - (f_2 E_2) f_1$ is orthogonal to E_2 (and, of course, E_3). Again, g will not be orthogonal to $E_1 = (a_{11}, a_{12}, a_{13})$ when the matrix (a_{ij}) is nonsingular. Then $\bar{g} = (x, y, z) = 1/(E_1 \cdot g)g$ will be a column vector such that $\bar{g} \cdot E_1 = 1$, $\bar{g} \cdot E_2 = 0 = \bar{g} \cdot E_3$. This procedure then gives us the first column of our inverse. A similar technique can be used to get the second and third columns. Our work is shortened by the fact, however, that in finding the second column for the inverse we find a vector orthogonal to E_3 and E_1 , but in our computations for the first column we already found a basis f_1, f_2 , equations (8), orthogonal to E_3 and thus a vector orthogonal to E_3 and E_1 can be written down at once, $h = f_1(E_1 f_2) - f_2(E_1 f_1)$. Then if h has the components (h_1, h_2, h_3) , then the second column of our inverse will be $(u, v, w) = (h_1/E_1 \cdot h, h_2/E_2 \cdot h, h_3/E_3 \cdot h)$. One obtains the third column of the inverse in the same manner as the first.

The procedure for inverting higher order matrices is essentially the same; the shortcut discussed above, namely of using information from previously solved columns to reduce work at a later stage, becomes much more prominent however.

5. An evaluation of the method.

For detailed information on and comparisons of various other methods for

inverting matrices we refer the reader to references 2, 3 and 4. We list here what we consider to be the important advantages and disadvantages of the vector method as compared with other methods.

(1) The vector method requires no special properties of the matrix (such as having the elements on the main diagonal large in comparison with the off-diagonal elements; the Crout method, for instance, has this requirement). (2) Perhaps, from the standpoint of its adaptability to machine operation, its chief merit is that it allows for frequent rescaling throughout the computation. We have encountered examples, in radiation analysis, of matrices where both the Crout and Gauss-Seidel methods failed on a CPC, the first because of scaling difficulties and the second because of extreme slowness of convergence. The vector method, however, was successful. (3) No division occurs except at the end. (4) It affords a convenient method for solving equations by hand. (5) For solving equations it requires less storage than the Crout method.

An important disadvantage is that it is slow in comparison with many other methods. It requires roughly $5n^3$ multiplications to invert an $n \times n$ matrix, whereas other direct methods are of the order of n^3 . In a subsequent paper we shall show how this can be obtained.

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